

Exercises — MDP, Return, Value Functions, Bellman Equations

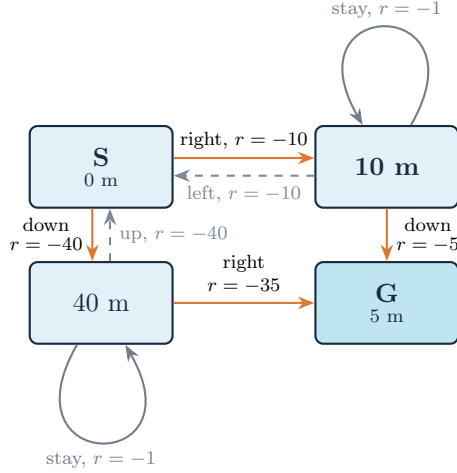
Lecture 1/4 — Reinforcement Learning

CentraleSupélec

Topics covered

- **Formalizing an MDP:** $\mathcal{S}$, $\mathcal{A}(s)$, $p(s', r \mid s, a)$.
- **Policies**, deterministic and stochastic, normalization condition.
- **Return** G_t and the role of the discount factor γ .
- **Value functions** v_π , q_π , and their relation.
- **Bellman equations**, expectation and optimality.
- **Proof** (Exercise 9): the effect of shifting all rewards by a constant.
- **Going further** (Exercises 10–11, from Sutton & Barto): the role of γ in choosing π_* , and the limits of the MDP framework.
- **Bonus booklet:** deriving the Bellman equations from the definition of G_t , the $v_*/q_*/\pi_*$ dictionary, three-argument notation, and episodic versus continuing tasks (originally Exercises 9, 12–14) now live in a separate booklet, updated each session — see `../bonus-booklet/`.

Reference environment (recall from lecture). Four states: S (start, 0 m), A (10 m), B (40 m), G (terminal, 5 m). Reward $r(s, a) = -|\text{altitude}(s') - \text{altitude}(s)|$: this is a **cost-of-effort** MDP (minimize total climbing effort to reach G), not a sparse goal-bonus one — there is no reward for arriving at G beyond the cost of the last step. $\gamma = 1$ unless stated otherwise. *Note:* the *stay* self-loop costs $r = -1$ (not free) — with $\gamma = 1$, a free ($r = 0$) self-loop would make an infinite stay tie with any finite-cost path to G , an unbounded-return degeneracy. At $r = -1$, lingering forever costs $-\infty$, so *stay* is always dominated and never appears in an optimal policy (see Exercise 7) — no special-casing needed.



Exercise 1 — Formalize the environment as an MDP

- List $\mathcal{S}$ and $\mathcal{A}(s)$ for each of the four states (careful: G is terminal, $\mathcal{A}(G) = \emptyset$).
- This environment is **deterministic**: for a given pair (s, a) , exactly one pair (s', r) has nonzero probability. Write $p(s', r \mid s, a)$ for the three pairs (S, right) , (A, stay) and (B, up) .
- Check that $\sum_{s', r} p(s', r \mid S, \text{right}) = 1$. Why does this sum always equal 1, regardless of the environment?

Exercise 2 — Two policies

Consider two policies on this environment:

- π_A (**deterministic**): always *right* at S , always *down* at A , always *right* at B .
- π_B (**stochastic**): at S , $\pi(\text{right} \mid S) = 0.3$ and $\pi(\text{down} \mid S) = 0.7$; elsewhere identical to π_A (deterministic at A and B).

- Write $\pi(a \mid S)$ for π_A and for π_B as a table (one row per possible action at S).
- Check $\sum_a \pi(a \mid S) = 1$ for both policies.
- Which of the two matches the “Dirac mass” mentioned in lecture (slide “Policy”) ? Write $\pi_A(\cdot \mid S)$ in that form.

Exercise 3 — Return and discount factor

Under π_A , the unique trajectory from S is

$$S \xrightarrow[R_1 = -10]{\text{right}} A \xrightarrow[R_2 = -5]{\text{down}} G \text{ (end)}.$$

- Compute the return G_0 (with $\gamma = 1$).
- Recompute G_0 with $\gamma = 0.9$.
- Compute G_1 (the return from state A onward) in both cases, and check the relation $G_0 = R_1 + \gamma G_1$.

- (d) In one sentence: how does γ change the relative weight of R_1 (immediate) versus R_2 (future) in G_0 ?

Exercise 4 — Value functions under π_A

Stay under π_A , with $\gamma = 1$.

- (a) By backward substitution from G (where $v_\pi(G) = 0$), compute $v_{\pi_A}(A)$ then $v_{\pi_A}(S)$.
- (b) Compute $v_{\pi_A}(B)$ (recall: π_A also goes *right* from B).
- (c) Compute $q_{\pi_A}(S, \text{right})$ and $q_{\pi_A}(S, \text{down})$.
- (d) Compute $q_{\pi_A}(A, \text{down})$ and $q_{\pi_A}(A, \text{left})$ (the latter leads back to S : use the value $v_{\pi_A}(S)$ found in (a)).
- (e) *Check*: compare $q_{\pi_A}(S, \text{right})$ and $q_{\pi_A}(S, \text{down})$ to the values $q_\pi(S, \text{right}) = -15$ and $q_\pi(S, \text{down}) = -75$ from the “Action-value function - example” slide in lecture. Is this consistent? Why (the lecture’s policy isn’t π_A)?

Exercise 5 — v_π as an average of q_π under π_B

- (a) Using the values of $q_{\pi_A}(S, \text{right})$ and $q_{\pi_A}(S, \text{down})$ found in the previous exercise (still valid under π_B : why?), and the probabilities $\pi(\cdot | S)$ of π_B (Exercise 2), compute $v_{\pi_B}(S)$ using

$$v_\pi(s) = \sum_{a \in \mathcal{A}(s)} \pi(a | s) q_\pi(s, a).$$

- (b) Is the result closer to $q_{\pi_A}(S, \text{right})$ or to $q_{\pi_A}(S, \text{down})$? Is this consistent with the 0.3 / 0.7 probabilities chosen?

Exercise 6 — Bellman expectation equation

- (a) Write the general Bellman expectation equation for $v_\pi(S)$ (sum over actions, then over (s', r)).
- (b) Instantiate it for π_B at S (two possible actions, deterministic environment so the sum over (s', r) has a single nonzero term per action).
- (c) Check that it reproduces $v_{\pi_B}(S)$ found in the previous exercise.

Exercise 7 — Optimal value function and optimal policy

Now A and B each have 3 actions available: the two moves already used, plus *stay* ($r = -1$).

- (a) For A and for B , compare all 3 available actions and justify that both *stay* and going back toward S are never optimal. Deduce $v_*(A)$ and $v_*(B)$.
- (b) Deduce $v_*(S)$ (compare *right* and *down* from S , using $v_*(A)$ and $v_*(B)$).
- (c) *Consistency check*: plug $v_*(S)$ back into the equations of (a) and check they still hold (fixed point of the optimality equation).

- (d) Deduce π_* . Compare to π_A (Exercises 4 and 5) and to $v_{\pi_A}(S)$ found in Exercise 4(a). What does the existence theorem for a deterministic optimal policy say about this result?

Exercise 8 — Returns from a reward sequence (classic, Sutton & Barto Ch. 3)

- (a) **Episodic task.** Suppose $\gamma = 0.5$ and the following sequence of rewards is received: $R_1 = -1$, $R_2 = 2$, $R_3 = 6$, $R_4 = 3$, $R_5 = 2$, with the episode ending at $T = 5$ (so $G_5 = 0$). Compute $G_0, G_1, \dots, G_5$. Work **backwards**, using $G_t = R_{t+1} + \gamma G_{t+1}$.
- (b) **Continuing task.** Now suppose $\gamma = 0.9$ and the reward sequence is $R_1 = 2$ followed by an **infinite** sequence of 7s (i.e. $R_2 = R_3 = \dots = 7$). Compute G_1 using the geometric series identity $\sum_{k=0}^{\infty} \gamma^k = \frac{1}{1-\gamma}$ (valid for $\gamma < 1$), then G_0 .
- (c) Episode (a) terminates after 5 steps; the reward sequence in (b) never ends. Why is G_1 still a **finite** number in (b)?

Exercise 9 — Do the signs of the rewards matter? (theory bonus, Sutton & Barto Ch. 3)

Every reward in this sheet's environment is **negative**, which may feel like an arbitrary modelling choice: why not add a constant to make them all positive, and maximize a “score” instead of minimizing a cost? This exercise settles the question — and the answer is not the same for continuing and for episodic tasks.

Fix a constant $c \in \mathbb{R}$ and define a new MDP, identical to the old one except that every reward is shifted: $R'_t = R_t + c$. Write v'_π for value functions in the shifted MDP.

- (a) **Continuing task**, $\gamma < 1$. Prove that there is a constant v_c , *independent of the state and of the policy*, such that

$$v'_\pi(s) = v_\pi(s) + v_c \quad \text{for every } s \text{ and every } \pi,$$

and give v_c in terms of c and γ . *Hint:* start from $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$ and use the geometric series of Exercise 8(b).

- (b) Deduce that the shift changes neither the ordering of policies nor π_* . Conclude: in a continuing task, only the **differences** between rewards carry information, not their signs.
- (c) **Episodic task**, $\gamma = 1$. Let T be the (random) time at which the episode ends. Show that now

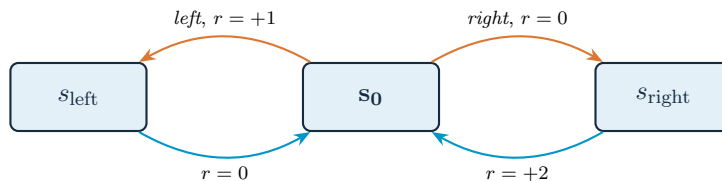
$$v'_\pi(s) = v_\pi(s) + c \mathbb{E}_\pi[T \mid S_0 = s],$$

and explain in one sentence why this is **not** the situation of (a).

- (d) **Counterexample on our own environment.** Take the environment of this sheet ($\gamma = 1$, *stay* available at A and B with $r = -1$) and shift every reward by $c = +11$. Compare the policy π_A ($S \rightarrow A \rightarrow G$, reaching G in 2 steps) with the policy that goes to A , plays *stay* k times, then goes to G . Compute both returns from S in the shifted MDP as functions of k , and conclude. What is $v'_*(S)$?
- (e) For which values of c does the shifted task stay well-posed? Relate your answer to the note on the *stay* self-loop at the top of this sheet ($r = -1$ rather than $r = 0$), and to Exercise 7(a).

Exercise 10 — The discount factor decides the optimal policy (Sutton & Barto 3.22)

Every environment so far has been **episodic**. Here is a **continuing** one — it never terminates. The only decision is taken in the top state s_0 , where two actions are available; every other transition is forced. Rewards are deterministic, shown on the arrows.



In s_{left} and s_{right} a single action is available (the return arrow below), so the agent decides only in s_0 : there are exactly two deterministic policies, π_{left} and π_{right} .

- (a) Write the infinite reward sequence seen from s_0 under each of the two policies.
- (b) Show that

$$v_{\pi_{\text{left}}}(s_0) = \frac{1}{1 - \gamma^2}, \quad v_{\pi_{\text{right}}}(s_0) = \frac{2\gamma}{1 - \gamma^2}.$$

Hint: each policy returns to s_0 every 2 steps, so group the rewards in pairs — the series is geometric with ratio γ^2 .

- (c) Which policy is optimal for $\gamma = 0$? For $\gamma = 0.9$? For $\gamma = 0.5$?
- (d) Solve $v_{\pi_{\text{left}}}(s_0) = v_{\pi_{\text{right}}}(s_0)$ exactly. What does this exercise say about γ that Exercise 3(d) did not?

Exercise 11 — Limits of the MDP framework (discussion, Sutton & Barto 3.2 and 3.3)

No computation — two questions worth a genuine discussion, with no unique answer.

- (a) Is the MDP framework sufficient to usefully represent *all* goal-directed learning tasks? Propose at least two clear counterexamples, saying in each case what exactly breaks.
- (b) Consider driving a car. The actions could be defined at the level of the accelerator, steering wheel and brakes; or one level below (the driver's muscles); or one level above (the choice of route). Where should the boundary between **agent** and **environment** be drawn? Is there a single right answer? What rule of thumb would you propose?

Notation & formulae

- **MDP:** $\mathcal{S}$ (states), $\mathcal{A}(s)$ (actions available at s), $p(s', r \mid s, a) \doteq \mathbb{P}[S_t = s', R = r \mid S_{t-1} = s, A_{t-1} = a]$.
- **Policy:** $\pi(a \mid s) \doteq \mathbb{P}[A_t = a \mid S_t = s]$, with $\sum_{a \in \mathcal{A}(s)} \pi(a \mid s) = 1$ (or $\int_{\mathcal{A}(s)} \pi(a \mid s) da = 1$ for continuous actions). Deterministic π : $\pi(\cdot \mid s) = \delta_{\pi(s)}$.
- **Return:** $G_t \doteq \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = R_{t+1} + \gamma G_{t+1}$.
- **Value functions:** $v_{\pi}(s) \doteq \mathbb{E}_{\pi}[G_t \mid S_t = s]$, $q_{\pi}(s, a) \doteq \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a]$.
- v_{π} as an average of q_{π} : $v_{\pi}(s) = \sum_{a \in \mathcal{A}(s)} \pi(a \mid s) q_{\pi}(s, a)$.
- **Bellman expectation equations:**

$$v_{\pi}(s) = \sum_a \pi(a \mid s) \sum_{s', r} p(s', r \mid s, a) [r + \gamma v_{\pi}(s')], \quad q_{\pi}(s, a) = \sum_{s', r} p(s', r \mid s, a) \left[r + \gamma \sum_{a'} \pi(a' \mid s') q_{\pi}(s', a') \right]$$

- **Optimal value functions:** $v_*(s) \doteq \max_{\pi} v_{\pi}(s)$, $q_*(s, a) \doteq \max_{\pi} q_{\pi}(s, a)$.
- **Bellman optimality equations:**

$$v_*(s) = \max_{a \in \mathcal{A}(s)} \sum_{s', r} p(s', r \mid s, a) [r + \gamma v_*(s')], \quad q_*(s, a) = \sum_{s', r} p(s', r \mid s, a) \left[r + \gamma \max_{a'} q_*(s', a') \right]$$

- **Optimal policy:** $\pi_*(s) \doteq \arg \max_{a \in \mathcal{A}(s)} q_*(s, a)$. At least one deterministic π_* always exists (finite MDP, bounded rewards).